

Bank Management System

1. Introduction

The Bank Management System is a console-based application developed using C++. The main purpose of this system is to perform basic banking operations efficiently using Object-Oriented Programming and File Handling concepts.

This application helps users to create accounts, deposit money, withdraw money, search customer details, display account information, and delete accounts while securely storing customer records in files.

2. Objectives

- To manage bank account records efficiently.
 - To perform banking operations digitally.
 - To reduce manual record maintenance.
 - To provide secure storage using file handling.
 - To learn OOP concepts using C++.
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3. System Features

Main Features

1. Create New Bank Account
 2. Display All Account Details
 3. Search Account by Account Number
 4. Deposit Money
 5. Withdraw Money
 6. Delete Account
 7. File Handling for Data Storage
 8. Menu Driven Console Application
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4. Technologies and Concepts Used

Technologies Used

- Programming Language: C++

- Platform: Console Application
- Compiler: Dev C++ / CodeBlocks

Concepts Used

- Classes and Objects
 - Functions
 - File Handling (`fstream`)
 - Loops (`while`, `do-while`)
 - Conditional Statements (`if`, `switch`)
 - Character Arrays
 - Input and Output Streams
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5. System Design

Class Design

Class: Bank

DATA MEMBERS	DESCRIPTION
name	Stores Account Holder Name
accNo	Stores Account Number
balance	Stores Account Balance

Functions Used

FUNCTION	PURPOSE
createaccount()	Creates new bank account
displayaccount()	Displays account details
Deposit Logic	Adds money to account
Withdraw Logic	Withdraws money from account
Search Logic	Searches account by number
Delete Logic	Deletes account record

6. Workflow

Step-by-Step Working Process

1. Program starts.

2. Main menu is displayed.
 3. User selects an option.
 4. Based on user choice:
 - Create account
 - Display accounts
 - Search account
 - Deposit money
 - Withdraw money
 - Delete account
 5. Data is stored/retrieved using files.
 6. Menu repeats until Exit option is selected.
 7. Program terminates.
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7. Sample Output

```
=====WELCOME TO BANK MANAGEMENT SYSTEM=====
```

```
1.Create an account
2.Display account details
3.Search account details
4.Deposit money
5.Withdraw money
6.Delete account
7.Exit
```

```
Enter choice: 1
```

```
Enter Account Holder Name: Ravi
Enter Account Number: 101
Enter Bank Balance: 5000
```

```
Account created successfully!
```

Deposit Example

```
Enter Account Number: 101
Enter Amount to deposit: 2000
```

```
Amount added successfully!
```

Withdraw Example

```
Enter Account Number: 101
Enter Amount to withdraw: 1000
```

```
Amount withdraw successfully!
```

Search Example

Enter Account Number to search: 101

Account Holder Name: Ravi

Account Number: 101

Bank Balance: 6000

8. Conclusion

This project is a beginner-friendly banking application that demonstrates the implementation of Object-Oriented Programming and File Handling concepts in C++. The system performs essential banking operations efficiently while maintaining persistent customer records securely.